

Structuring of a Predator Population Through Temperature-Mediated Effects on Prey Availability¹

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ADAMS, S. M., R. B. MCLEAN, AND M. M. HUFFMAN. 1982. Structuring of a predator population through temperature-mediated effects on prey availability. *Can. J. Fish. Aquat. Sci.* 39: 1175–1184.

Temperature can control the structure of a predator population by regulating the abundance and size availability of prey. Relatively small differences in winter temperatures between years can have a major influence on largemouth bass (*Micropterus salmoides*) population structure because of the threshold relationship between temperature and the lower lethal limits of shad. The relationship between prey availability and predator growth and mortality was established through field measurements of consumed energy. Growth and consumption of largemouth bass in Watts Bar Reservoir, Tennessee, was lower in 1979 than in 1980 as a result of lower winter water temperature in 1979, which reduced the abundance and availability of adequate-sized prey. Depending on the severity of the winter, bass that do not attain 25 ± 5 cm by the end of their first growing season do not survive to annulus I formation because of the unavailability of appropriate-sized prey in the spring and the inability of small bass to store sufficient energy reserves. Selective mortality favoring survival of larger individuals in an age-class is operating in the Watts Bar largemouth bass population. This has significant implications for predator growth in ecosystems that experience large seasonal fluctuations in prey availability.

Key words: predator population structure, prey availability, reservoir temperature, energy consumption, growth, largemouth bass, selective mortality, shad

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La température peut contrôler la structure d'une population de prédateurs par régulation de l'abondance et de la disponibilité de taille des proies. Des différences relativement faibles des températures hivernales d'une année à l'autre peuvent influencer beaucoup sur la structure des populations d'achigans à grande bouche (*Micropterus salmoides*) par suite de la relation de seuil qui existe entre la température et les limites létales inférieures de l'alse savoureuse. Des mesures sur le terrain de l'énergie consommée ont permis d'établir la relation entre la disponibilité des proies et la croissance et la mortalité des prédateurs. La croissance et la consommation d'achigans à grande bouche dans le réservoir de Watts Bar, dans le Tennessee, étaient plus faibles en 1979 qu'en 1980 par suite d'une température de l'eau en hiver plus basse en 1979, qui causa une diminution de l'abondance et de la disponibilité de proies de taille convenable. Selon la sévérité de l'hiver, les achigans qui n'ont pas atteint 25 ± 5 cm à la fin de leur première saison de croissance mourront avant formation de l'annulus I parce qu'il n'y a pas de proies de taille convenable au printemps et que les petits achigans ne peuvent stocker des réserves énergétiques suffisantes. Il y a dans la population d'achigans à grande bouche de Watts Bar une mortalité sélective qui favorise la survie des grands sujets d'une classe d'âge. Ceci a d'importantes implications sur la croissance des prédateurs dans des écosystèmes où la disponibilité des proies accusent de fortes fluctuations saisonnières.

Received August 19, 1981

Accepted April 13, 1982

Reçu le 19 août 1981

Accepté le 13 avril 1982

¹Research sponsored by the Office of Health and Environmental Research, U.S. Department of Energy, under contract W-7405-eng-26 with Union Carbide Corporation. Publication No. 1765, Environmental Sciences Division, Oak Ridge National Laboratory.

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TEMPERATURE can influence the dynamics of fish populations through control of physiological processes (Fry 1971), control of behavioral thermoregulation (Coutant 1977), interaction with toxicants (Cairns et al. 1975), effects on spawning and survival of early life stages (Summerfelt 1975), and promotion of disease (Reichenbach 1974). Temperature also may exert a regulating influence on the structure of predator populations by controlling the density, size availability, and quality of prey available to a predator. In many large freshwater systems of North America, threadfin shad (*Dorosoma petenense*) and gizzard shad (*Dorosoma cepedianum*), the primary forage of several predators experience cold-stress mortality when water temperatures drop below 9°C, with complete mortality occurring around 4°C (Griffith 1978; McLean et al. 1980). When mass winter mortalities (Hassler 1953; Parsons and Kimsey 1954; McLean et al. 1980) or even partial mortality of forage populations occur, predators are usually faced with a shortage of their preferred food during the spring and early summer. Response of predator populations to severe seasonal or yearly fluctuations in prey abundance is not well documented. In northern systems, reduction in growth rate and fecundity of sauger (*Stizostedion canadense*) and walleye (*Stizostedion vitreum vitreum*) has been attributed to winter kill of gizzard shad (Forney 1965; Nelson 1969). Minton et al. (1981) stated that sauger in a southeastern reservoir cope with large seasonal variations in threadfin shad availability by increasing their consumption (over prewinter levels) of vulnerable cold-stressed shad in early winter and storing this excess energy in large fat deposits which are metabolized in the spring and early summer when shad are scarce.

The objective of this study was to determine if temperature influences the size structure of a predator population by affecting the size of the predator's preferred prey. We measured growth, mortality, and food consumption of largemouth bass (*Micropterus salmoides*) as possible variables which are affected by both a fluctuating prey base and size availability of prey. We chose largemouth bass for this study because of its importance as a game fish throughout North America and our hypothesis that a fluctuating prey base is an indirect determinant of bass growth and survival. This is the first study on adult largemouth bass where the effects of temperature on population structure, mediated through prey availability, have been investigated and where energy consumption has been measured for natural populations.

Methods

This study was conducted in Watts Bar Reservoir, a 13 300-ha impoundment of the Tennessee River in eastern Tennessee. The reservoir, characterized by shallow flats about the main channel which stratifies during the summer, is a mesotrophic system with a diversity of native and introduced fish species. Threadfin and gizzard shad are the dominant forage species in the reservoir, while largemouth and smallmouth bass (*Micropterus dolomieu*), sauger, striped bass (*Morone saxatilis*), and white bass (*Morone chrysops*) are the principal predators. The shad populations, and particularly the threadfin, undergo dramatic temporal fluctuations in abundance, due primarily to winter cold-kill (McLean et al. 1980).

We collected largemouth bass biweekly in 1979 and 1980 by electroshocking with equal effort in brush piles, submerged vegetation, and rocky shores. Electrofishing is a relatively nonselective method of sampling all size-classes of largemouth bass in a population (Woodrum 1974; Frankenberg 1960). A minimum of 35 bass (ages 1+ and older) were captured each month, except when unfavorable water temperatures (<10°C and >27°C) forced bass into deep water. Fish were frozen immediately with dry ice to stop digestion. Prey species also were collected, by electroshocking, for caloric and ash analysis.

Total lengths, weight, and viscera weights were recorded, scales removed, and stomach contents analyzed for all bass. Two bass from each age-group were saved each month for caloric and ash analysis of whole body and viscera. Lengths (total, standard, or backbone) and weights of ingested prey were recorded and partially digested prey were identified by the presence of gizzards, otoliths, or dorsal spines.

We determined the caloric content of whole bass (carcass, viscera, and gonads), viscera, and representative prey each month with a Parr adiabatic bomb calorimeter following standard methods (Paine 1971). Samples were freeze-dried, ground into a homogeneous mixture with a Wiley mill, and two subsamples combusted from each sample. To determine sample replicability, seven replicates each of samples with high ash (27%) and low ash (8%) were analyzed yielding caloric values of 4.028 ± 0.063 and 6.487 ± 0.061 cal·g⁻¹, respectively. Because of this precision, we subsequently ran only two determinations per sample. Ash contents were determined from duplicate subsamples of fish tissue ignited for 18 h at 500°C.

For digestion rate studies, field-collected 400 to 700-g bass were individually placed in 1-m³ vexas mesh cages in large circular tanks supplied with running well water with computer-controlled temperatures. Bass were held in the laboratory for 6 wk before the digestion experiments, and were fed large fathead minnows (*Pimephales promelas*). Digestion rate experiments were run at 14, 17, 20, 23, 26, and 28°C, spanning the range of temperatures at which bass were observed to feed in the field. We did not feed bass 36 h prior to experiments, and experiments were conducted 2 wk apart to allow normal feeding to resume. For each experiment, 25–30 fish were each fed 2% of their body weight (mean field daily ration determined by consumption measurements). Only bass that fed voluntarily were used in the experiments, because force-feeding has been shown to bias digestion rates (Swenson and Smith 1973). At various intervals following feeding, and depending on temperature, three to four bass were anesthetized with MS-222 and their stomachs gently pumped (Seaburg 1957). We calculated time to 95% digestion at each temperature by a regression of the fraction digested at each sample interval against hours following feeding. Time to 95% digestion was considered the endpoint of digestion because the remaining 5–10% is considered the residual phase where digestion of hard parts is slowed significantly (Swenson and Smith 1973). Digestion rate of live threadfin shad was also conducted at two temperatures and the time to 95% digestion compared to minnows using Student's *t*-test and the Satterthwaite (1946) procedure to approximate the degrees of freedom.

We calculated monthly consumption or daily ration of bass in the field by the relationship

$$C = \frac{\sum_{i=1}^n (Pw_i / Bw_i)}{n},$$

where: C = consumption or daily ration ($\text{mg} \cdot \text{g}^{-1} \cdot \text{d}^{-1}$), Pw_i = weight of prey ingested during a defined 24-h period, Bw_i = weight of bass that consumed prey Pw_i , and n = number of bass in sample including those with empty stomachs. Monthly ration estimates ($\text{mg} \cdot \text{g}^{-1} \cdot \text{d}^{-1}$) were calculated for each age-class (I^+ – IV^+), converted to consumed energy ($\text{cal} \cdot \text{g}^{-1} \cdot \text{d}^{-1}$) by accounting for prey caloric value, and these monthly values summed to yield annual rations for each age-class.

Prey ingested during 24 h was determined from the digestion rate of prey given the temperature and time of bass capture, original weight of an ingested prey, and time of prey ingestion. Temperature was taken in each general area in which bass were captured. Digestion rate of prey at any reservoir temperature was determined from a regression of time to 95% digestion at the six temperatures tested in the laboratory. We calculated original weights of ingested prey from a series of backbone length, standard length, and total length regressions (Minton et al. 1981). Time of ingestion was determined by backcalculating, from the time of capture, the number of hours that the prey had been in the stomach based on stage of digestion. Stage of digestion (proportion digested) was calculated as $1 - (\text{digested weight}/\text{original weight})$. A fish was given credit for consuming prey only if the prey was ingested within a designated period which included 24 h pre-captus to the capture of the last bass. When digestion rate was greater than 24 h, food in the stomach was a record of ingestion 24 h prior to capture. If digestion rate was less than 24 h, we sampled twice (12 h apart) within a 24-h period and combined fish from both sampling periods to account for those bass that fed within the 24 h but may have digested their meal before capture.

We determined age of bass by analyzing scales taken from the body near the tip of the left pectoral fin. Lengths at each annulus were backcalculated by substituting the intercept values of the total length–anterior radius relationship into the expression

$$BL = I + (S'/S)(TL - I),$$

where: BL = backcalculated length at annulus, I = intercept of the length–scale radius equation, S' = annulus distance, S = scale radius, and TL = total length of fish. The intercept (I) was calculated by a geometric means regression procedure of Ricker (1973) which minimized both the vertical and horizontal regression sum of squares. Weight-specific caloric growth ($\text{cal} \cdot \text{g}^{-1} \cdot \text{d}^{-1}$) for a month was calculated as the mean weight-specific caloric content of the whole fish ($\text{cal} \cdot \text{g}^{-1}$) during that month times the mean weight gain over the month divided by the monthly mean weight.

Prey size–frequency distribution over the year was determined from biweekly gillnetting data of Minton et al. (1981) collected from April 1979 to March 1980. Weekly impingement data were collected by McLean et al. (1980) from the intake screens of Kingston Steam Plant on Watts Bar

Reservoir during the winters of 1976–80. Biweekly surface and mid-depth (3 m) temperatures were taken at several areas of the reservoir, including cove and main reservoir stations.

Results

TEMPERATURE

The winter of 1979 was colder than the winter of 1980. During January and February of 1979, temperatures averaged 4–5°C with one week in February reaching 2.9°C. During the winter of 1980, temperatures ranged between 7 and 10°C, except for one week that averaged 6.4°C. Summer water temperatures were also higher in 1980, with highs reaching 30°C for two weeks in August 1980 and 27.5°C in August 1979.

GROWTH AND SIZE CHANGES

Caloric growth and growth in size of bass were generally greater in 1980 than in 1979 (Table 1). Caloric gain of an average age I^+ and II^+ bass was 3.5 and 1.7 times greater, respectively, in 1980 than in 1979 while caloric growth of age III^+ was about the same both years. The average weights at the end of the growing season were 1.8, 1.4, and 1.2 times greater for ages I^+ , II^+ , and III^+ , respectively, in 1980 compared to 1979.

Backcalculation to determine size at each annulus formation for various year-classes of bass revealed that backcalculated lengths were larger than observed lengths for each respective age. There is a consistent and progressive increase in backcalculated lengths at each annulus beginning with the 1979 year-class and proceeding back to the 1976 year-class (Table 2). For example, the length at annulus I for fish caught in 1980 was 23.5, 25.5, 27.2, and 29.2 cm for the 1979, 1978, 1977, and 1976 year-classes, respectively. This pattern is also evident for lengths at annulus II and III (Table 2). Backcalculated lengths at annulus I ranged from 21.9 to 29.2 cm (Table 2), but the observed lengths for 0^+ bass at the end of their first growing season ranged from 7.5 to 29.3 cm ($\bar{x}$ = 16.1 cm, Fig. 1A). The average size of 0^+ bass shifted from 16.1 cm in early December 1980 at the end of the growing season to 22.9 cm in the early spring of 1981 (April) prior to annulus formation (Fig. 1B).

DIGESTION AND CONSUMPTION

The time required for 95% digestion decreased as temperature increased (Fig. 2). Digestion rate of shad was not significantly different from digestion rate of minnows at 26°C ($P > 0.10$), but at 20°C digestion rate of minnows was slightly faster ($0.10 > P > 0.05$) than shad. This small difference in digestion at 20° was assumed to have no effect on the final consumption estimates given all the other sources of inherent errors that might be involved in estimating consumption of fish in the field such as effects of meal size and type, predator size, and frequency of feeding on digestion rate (Swenson and Smith 1973). Digestion rate was a linear function of time at each temperature (excluding the residual

TABLE 1. Annual growth, consumption, and gross growth efficiency (K_1) of each major age-class of largemouth bass in Watts Bar Reservoir in 1979 and 1980.

Year caught/ age-class	Mean length (cm) $\pm$ SD		Mean weight (g)		No. fish	Mean weight over year(g) ^b	Annual weight gain(g)	Weight- specific caloric content (cal/g)	Total caloric gain (cal/g) ^c	Consump- tion (cal/g)	K_1
	Initial ^a	Final ^a	Initial	Final							
1979											
I ⁺	23.1 $\pm$ 1.9	26.8 $\pm$ 3.1	187.6	245.0	42	216.3	57.4	1,140	302.5	1526	0.20
II ⁺	29.9 $\pm$ 2.3	35.6 $\pm$ 2.1	422.0	677.8	71	549.9	255.8	1,180	547.8	2743	0.20
III ⁺	35.9 $\pm$ 3.2	43.2 $\pm$ 2.2	758.5	1352.7	53	1055.6	594.2	1,390	783.0	4738	0.17
1980											
I ⁺	23.0 $\pm$ 1.5	30.8 $\pm$ 3.3	153.4	438.8	63	296.1	285.4	1,100	1060.2	4780	0.22
II ⁺	31.4 $\pm$ 3.6	38.9 $\pm$ 2.8	448.6	962.7	118	705.7	514.1	1,280	937.8	4754	0.20
III ⁺	38.5 $\pm$ 3.3	45.9 $\pm$ 1.8	902.7	1675.3	72	1289.0	772.6	1,370	821.1	5015	0.16
IV ⁺	45.9 $\pm$ 2.5	51.6 $\pm$ 2.7	1660.8	2472.4	22	2066.6	811.6	1,430	561.6	4917	0.11

^aInitial = start of growing season, final = end of growing season.^bLinear growth between monthly measurements assumed.^cCalculated as weight-specific caloric content $\times$ weight gain/mean weight.

TABLE 2. Backcalculated length at each annulus for the 1976–79 year-classes of largemouth bass in Watts Bar Reservoir.

Year-class/ year caught	Age when caught (yr)	No. fish	Total length, cm ($\pm$ SD), at annulus			
			1	2	3	4
1976						
1979	3	16	28.6(3.1)	36.2(2.9)	40.1(2.9)	
1980	4	22	29.2(3.3)	37.3(2.0)	43.3(1.7)	46.6(2.2)
1977						
1979	2	51	25.4(3.1)	31.2(3.2)		
1980	3	75	27.2(3.2)	34.8(2.8)	39.0(3.2)	
1978						
1979	1	45	21.9(2.3)			
1980	2	121	25.5(3.0)	31.8(3.2)		
1979						
1980	1	54	23.5(3.1)			

phase), while time to 95% digestion was an exponential function of temperature (Fig. 2) as found for other predators such as pike (Diana 1979), walleye, and sauger (Swenson and Smith 1973).

An average individual of age-class I⁺ and II⁺ consumed 62 and 42% fewer calories, respectively, in 1979 than in 1980, while consumption of age III⁺ was about equal both years (Table 1). Even though consumption of all prey was less in 1979, temporal consumption patterns were similar (Fig. 3). Gizzard shad dominated the diet both years, particularly in the spring before the smaller threadfin were available. In both years, feeding on threadfin shad increased in late spring when spawning shad were congregating in coves, and in the fall when 0⁺ threadfin became available. Young threadfin shad dominated the diet of bass in the fall of 1980 (Fig. 3), while in the preceding fall, intake of all prey was low. Alternative prey such as bluegill (*Lepomis macrochirus*) and freshwater drum (*Aplodinotus grunniens*) contributed only 15% of the total energy intake until the summer and early fall of both years when they made up 20–25% of the total.

Size of ingested prey varied between years and seasons. Ages I⁺ and IV⁺ bass ingested similar-sized prey both years, while ages II⁺ and III⁺ consumed larger prey in 1980 (Table 3). In the winter–spring periods large shad (15.0–17.5 cm) made up 35% of the diet in 1979 and only 26% in 1980. During the summer of 1979, 37% of the diet were the intermediate 12.0- to 15.0-cm shad, while in 1980 this group contributed only 20% of the diet. Young (0⁺) threadfin shad (5–10.0 cm) were important in the diet only during the fall of both years. The reduced consumption by the bass population in 1979 is reflected in a higher percentage of empty stomachs, a lower number of shad per bass stomach, and ingestion of smaller prey by the II⁺ and III⁺ age-classes (Table 3).

PREY AVAILABILITY

The amount of energy consumed is influenced by prey availability where availability is expressed as abundance and

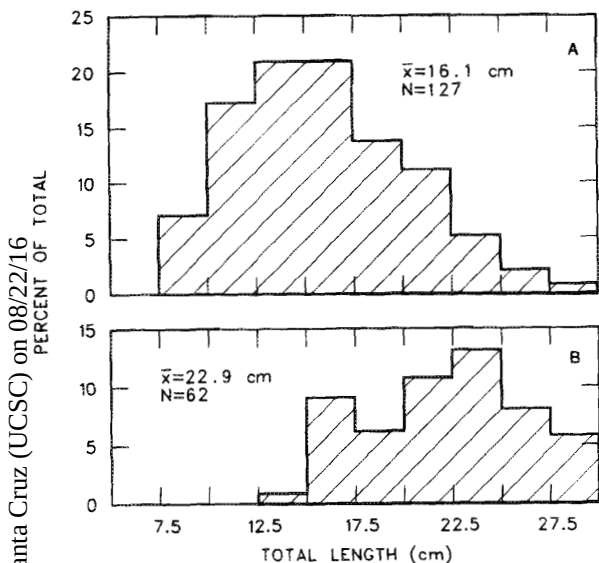


FIG. 1. Shift in length-frequency (2.5-cm size groups) of the 0⁺ largemouth bass collected in December 1980 at the end of the growing season (A) and in the early spring of 1981 prior to annulus formation (B).

the distribution. Relative abundance of prey was inferred from predator food habits and seasonal temperature structure in the reservoir, while size structure of the shad population in the reservoir was reflected by gill net and impingement data. The distribution of shad sizes captured in the gill nets and ingested by bass were similar each season. Ingested prey size, therefore, reflects size availability in the reservoir. Sizes of prey available to bass in the late winter and spring appear to depend on minimum winter water temperature (Fig. 4). Threadfin shad and small gizzard shad are stressed by cold and become disoriented around 9°C (Griffith 1978). When shad become cold-stressed or die they can become impinged on the screens of water-intake structures such as power plants (McLean et al. 1980). The size distribution and numbers of shad impinged at different temperatures give a

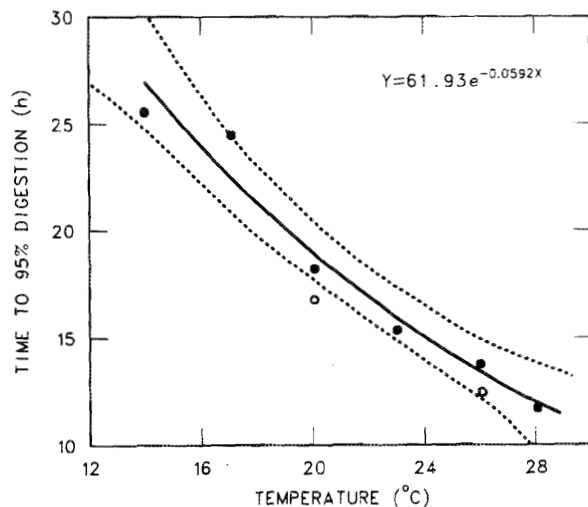


FIG. 2. Relationship between time to 95% digestion (h) and temperature for largemouth bass fed shad (open circles) and minnows (closed circles). The 95% confidence limits for the regression curve are shown.

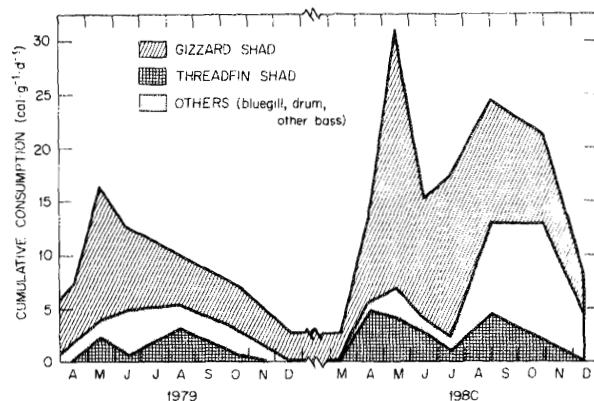


FIG. 3. Temporal consumption of prey by largemouth bass in Watts Bar Reservoir over two growing seasons.

TABLE 3. Mean length ($\pm$ SD) and weight of prey ingested by each age-class of largemouth bass and feeding data for all age-classes combined during 1979 and 1980 in Watts Bar Reservoir.

Year caught/ age-class	Mean length of ingested prey (cm $\pm$ SD)	Mean wt. of ingested prey (g)	% empty stomachs	Mean no. shad/bass	Mean no. prey/bass	Consumption (mg $\cdot$ g ⁻¹ $\cdot$ d ⁻¹)
1979						
I ⁺	8.4 $\pm$ 2.1	6.2				
II ⁺	10.2 $\pm$ 3.1	9.0				
III ⁺	14.7 $\pm$ 3.5	27.1				
IV ⁺	16.9 $\pm$ 5.1	41.3				
I ⁺ –IV ⁺			65	0.37	0.47	0.88
1980						
I ⁺	8.3 $\pm$ 2.2	6.0				
II ⁺	13.4 $\pm$ 2.7	20.5				
III ⁺	15.6 $\pm$ 2.8	32.5				
IV ⁺	16.5 $\pm$ 4.0	38.4				
I ⁺ –IV ⁺			54	0.56	0.65	1.70

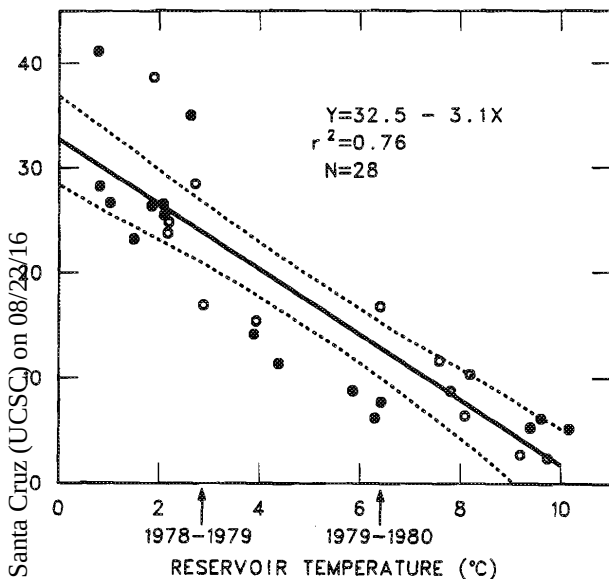


FIG. 4. Relationship of reservoir temperature and average weight of shad impinged on the intake screens of Kingston Steam Plant during the winter of 1976-77 (closed circles) and the winter of 1977-78 (open circles). Arrows indicate weekly average minimum water temperatures in the winter of 1978-79 and 1979-80. The 95% confidence limits of the regression are shown.

relative indication of the sizes removed by cold kills and the relative magnitude of cold kills on the shad population in the reservoir (McLean et al. 1980). A regression of mean weight of shad impinged against reservoir temperature indicated that a linear ($r^2 = 0.76$) was better than a nonlinear fit ($r^2 = 0.73$). The average size of shad stressed by low temperature increased as water temperature decreased, with larger shad killed mainly below 3°C (Fig. 4). In the winter of 1978-79, the lowest weekly minimum temperature was 2.9°C, and only those shad larger than 23.5 g (13.3 cm) probably survived. In the relatively mild winter of 1979-80, the lowest weekly average temperature was 6.4°C, and most shad larger than 12.6 g (10.7 cm) should have survived the winter (Fig. 4).

PREDATOR CONDITION

The visceral-somatic index (% body weight as viscera) and weight-specific caloric content of the viscera were dependent on largemouth bass size (Fig. 5, 6). As predator size increased, the visceral-somatic index increased rapidly, then leveled off (Fig. 5). Viscera weights in the fall tended to be higher than in the spring, especially for those bass on the right side of the inflection point in Fig. 5. In both seasons, the inflection point was about 700 g which approximates the size of an average age II⁺ bass. The weight-specific caloric content of the viscera also increased with bass size, indicating that larger fish have relatively higher lipid stores in the viscera (Fig. 6). Age-class I⁺ had significantly lower ($P < 0.05$) viscera weights each season, while differences between the other ages were not significant.

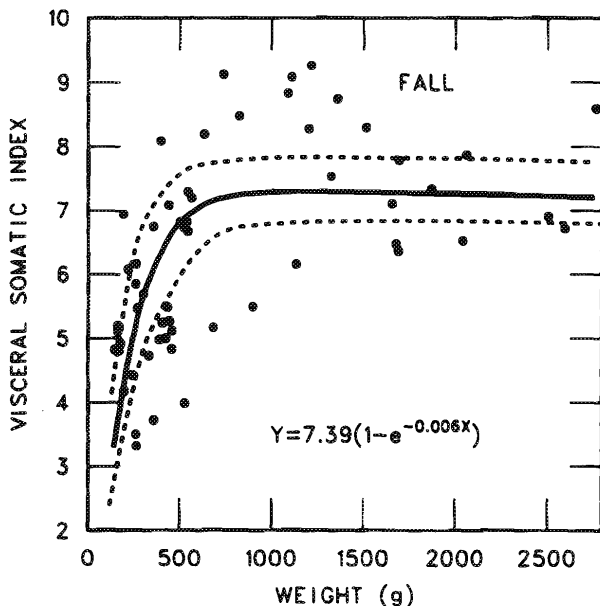
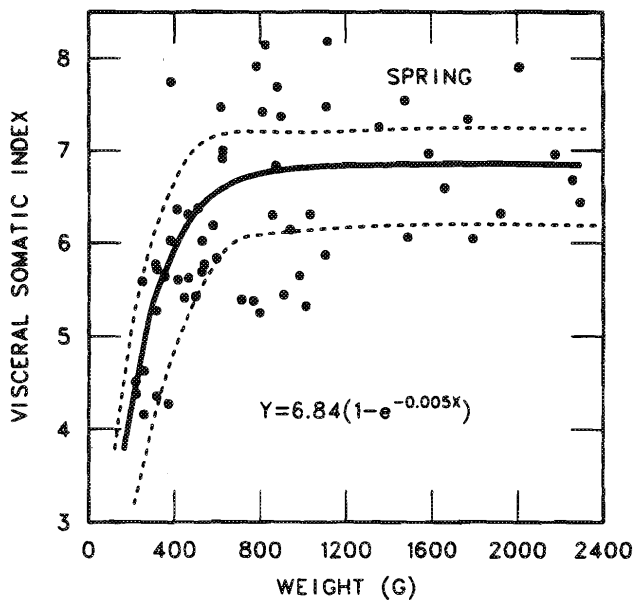


FIG. 5. Relationship between percent of largemouth body weight as viscera and total bass weight during the spring and fall of 1980 with 95% confidence limits of the regression.

Discussion

Relatively small differences in winter water temperature between years (2.9°C vs. 6.4°C) could result in major effects on predator populations. These effects may occur as a result of the threshold relationship between temperature and the lower thermal lethal limits of prey. Field measurements of consumption allowed us to establish the relationship between prey availability and predator growth and mortality.

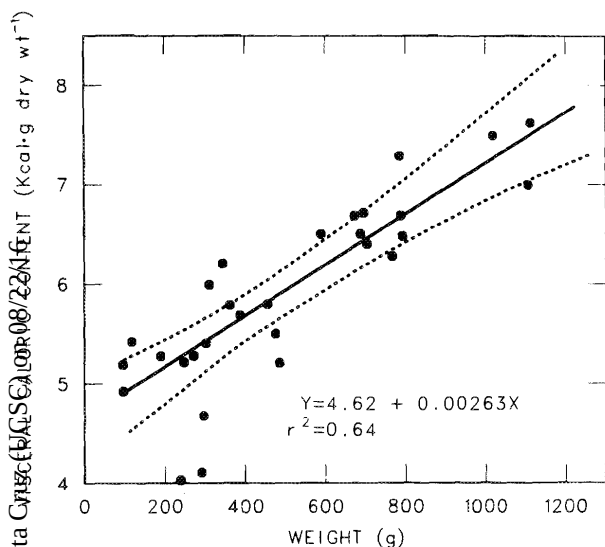


FIG. 6. Relationship between weight of largemouth bass caught during all seasons and caloric content of viscera. The 95% confidence limits of the regression line are shown.

CONSUMPTION AND GROWTH

The consumption methodology developed in this study is designed for warmwater predators that feed at least once a day during all seasons and caloric content of viscera only. The field consumption models of Diana (1979) for northern pike (*Esox lucius*), Popova and Sytina (1977) for pike-perch (*Stizostedion lucioperca*), and Swenson and Smith (1973) for walleye are designed for cold-water species that generally feed less than once a day and have digestion rates longer than 1 d. The consumption methodology appears to produce realistic estimates of daily ration that resulted in gross conversion efficiencies (K_1) ranging from 0.22 for age I⁺ bass to 0.11 for age IV⁺ (Table 1). This trend of decreasing conversion ratios with increasing age appears reasonable and is consistent with field and laboratory values (6–29%) for predators of similar sizes and life styles (Backiel 1971; Swenson and Smith 1973; Minton et al. 1981; Niimi 1981). Also, three separate estimates of consumption taken 1 wk apart in May 1980 yielded daily ration values of 29.5, 31.1, and 32.5 cal · g⁻¹ · d⁻¹, suggesting replicability of the method. Reduced consumption by the bass population in 1979 was probably related to low abundance of all shad smaller than 3.3 cm, particularly in the late winter and spring. Lower consumption rate, larger number of empty stomachs, and smaller number of shad and other prey per bass stomach during 1979 indicate that prey abundance was probably lower that year. Swenson (1977) found that consumption rates of prey in northern lakes were highly correlated with density of prey. Lewis et al. (1974) suggested that percentage of empty stomachs might be a means of evaluating the availability of forage in lakes, and Zweigacker and Summerfelt (1973) found an inverse relationship between percent empty stomachs and growth for largemouth bass.

Low water temperatures in early 1979 (<5°C) reduced the abundance of those shad small enough to be consumed by age

I⁺ bass (<30 cm) and small age II⁺ bass (<31 cm) (Fig. 4, Table 4). Consequently, consumption by these small-sized bass was depressed until midsummer when smaller prey became available. During the entire 1979 growing season age II⁺ bass were also consuming prey much smaller than the maximum size they are capable of ingesting (Table 3), possibly resulting in an energetic penalty for ingesting less than optimal size prey (Werner 1979). Growth, consumption, and size of prey ingested were about the same both years for age III⁺ bass, indicating sufficient numbers of appropriate-sized prey in 1979. These bass consumed the larger shad which were less affected by the low temperatures during both winters (Fig. 4).

The slight depression in consumption and growth during periods of high prey abundance in the summer can be partially explained in terms of largemouth bass thermal preference. The final thermal preferendum of largemouth is between 25 and 27°C (Coutant 1975) and, when surface temperatures were greater than 27°C (August–mid-September), bass inhabited deeper, cooler water (3–5 m). Shad were observed feeding in the upper parts of the water column during the summer, which might spatially segregate them from bass. Bass demonstrated an increase in ingestion of demersal prey such as freshwater drum during this time (Fig. 3). Bluegill, another alternate prey, contributed less than 10% of the total energy ingested by bass during both summers. The thermal preference of bluegill is also higher than that for bass (Reynolds and Casterlin 1976), which would help explain low consumption of this prey. Sunfish appear to be less vulnerable than shad to predation (Lewis and Helms 1964; Aggus 1973), possibly because of decreased efficiency of prey capture in structures such as brush piles that are inhabited by bluegill (Glass 1971).

MORTALITY AND SURVIVAL

The disappearance of the smaller individuals from the largemouth bass population in Watts Bar Reservoir during the winter was due to either growth or mortality of these small fish. Evidence indicates that largemouth bass in this reservoir do not feed or grow during the winter. First, prey was relatively scarce during the winter for those bass smaller than 20 cm (Table 4, Fig. 4). Only 2% of the total shad captured in gill nets during the winter–spring were small enough (8.3 cm) for 20-cm bass to swallow. Also, only 4 bass of the 48 collected during the winter (mid-December to mid-March) contained prey, and these bass were captured in thermal coves of the reservoir where stream input maintained cove temperatures 3–6°C above the main reservoir (McLean et al. 1980). Second, bass are not active and do not feed or grow in temperatures of 10°C or less (Warden and Lorio 1975; Johnson and Charlton 1960; Keast 1968), and reservoir temperatures were 10°C or less from mid-December to mid-March. Third, the condition factor of bass declined during the winter–spring indicating that bass of a fixed length were losing weight during this time of low metabolism. Adams et al. (1982) reported that an average II⁺ female and male largemouth bass in Watts Bar reservoir decreased in weight from 755 to 651 g and from 596 to 551 g, respectively, during the winter. The most convincing evidence supporting no winter growth is that

TABLE 4. Maximum lengths of shad that largemouth bass of various sizes can swallow, and the percentage of these maximum length shad caught in gill nets and found in bass stomachs during 1979.

Bass length (cm)	Maximum length shad bass can ingest (cm) ^a	% of total shad in gill nets ≤ maximum prey length			% of total shad in bass stomachs ≤ maximum prey length		
		Winter— spring	Summer	Fall	Winter— spring	Summer	Fall
15	6.5	0	1	1	0	3	15
20	8.3	2	3	8	1	8	28
25	10.4	5	11	25	7	20	48
30	12.7	15	33	38	33	52	63
35	15.3	76	84	69	73	88	86
40	18.3	85	98	97	95	95	97
45	21.4	100	100	100	100	100	100

^aMaximum shad lengths based on the relationship of Jenkins and Morais (1976): $y = 0.0141x^2 + 0.166x + 1.0772$, where y = prey size (in.) and x = predator size (in.) (1 in. = 25.4 mm).

no scale growth occurred on any bass from December until mid-April when water temperatures rose above 10°C, and bass were feeding.

Because largemouth bass do not grow during the winter, disappearance of the smaller individuals must be due to selective mortality. The progressive increases in backcalculated lengths (negative Lee's phenomenon) at each annulus with older age-classes (Table 2) indicate that selective mortality and not year-to-year variation in growth are primarily responsible for the distribution of sizes within an age-class. If annual fluctuation in growth affected backcalculated sizes, then this would be reflected in the lack of a consistent pattern of size increases at each annulus with each older age-class (random size distribution instead). Negative Lee's phenomenon "is certain to have been caused by selective mortality of one sort or another" (Ricker 1969).

To survive through the spring stress period of reduced prey availability, the predator must have either adequately sized prey or sufficient energy reserves. Evidence was presented earlier that severity of the winter dictates the sizes of prey available in the spring. Even in relatively mild winters, smaller shad (<9 cm) probably suffer mortality, which significantly reduces the forage base for bass smaller than 20 cm (Table 4). The low viscera somatic index (Fig. 5) and weight-specific caloric content of viscera (Fig. 6) also indicate that these smaller fish store very little fat. Therefore, the combination of low fat stores and no available food in the spring results in overwinter mortality of fish less than 20 cm. Other studies support this conclusion. Oliver et al. (1979) found that small smallmouth bass had less chance of surviving simulated winter temperatures in the laboratory because of the higher metabolic demands of small bass which resulted in an exhaustion of their energy supply. Shuter et al. (1980) also reported a higher rate of winter mortality among small young-of-the-year smallmouth bass because the larger fish had a relatively larger supply of energy reserves. First-year mortality in winter was found by Toney and Coble (1979) to be higher for smaller largemouth bass than larger bass. In contrast, Saiki and Tash (1978) reported that 25- to 35-cm largemouth bass in a backwater area of the Colorado River starved to death in

the spring because threadfin shad in this system were either too small (2–5 cm) or too large (16–19 cm) to be consumed.

Size-selective mortality of smaller individuals in a predator population seems to occur in systems where threadfin and gizzard shad are the primary forage and where these prey experience annual winter die-offs. Age-growth data for largemouth bass in Bull Shoals Reservoir, Arkansas (Bryant and Houser 1971), and for sauger and striped bass in Watts Bar Reservoir (Minton et al. 1981; Higginbotham 1979) indicate selective mortality of smaller fish under a fluctuating prey base. In contrast, bass feeding primarily on a stable prey base of bluegills in an Alabama reservoir (Olmsted 1974) and in a Wisconsin lake (Mraz and Threinen 1955) showed no evidence of size-selective mortality. No size-selective mortality was observed for bass in a California lake where threadfin shad are abundant all year (Von Geldern and Mitchell 1975).

Size-selective mortality results in larger average-sized bass of the same age in the spring compared to the previous fall. For example, the average length of age I⁺ bass was 26.8 cm at the end of the 1979 growing season and 31.4 cm the following spring at the time of annulus II formation (Table 1). Only the average size of the 0+ age-class at the end of the growing season can be compared directly to annulus I lengths of bass from other systems, because these young fish have not yet experienced the size bias attributed to winter mortality.

To increase the probability of survival through their first winter, therefore, 0+ bass in Watts Bar Reservoir must attain 25 ± 5 cm by mid-December. A severe winter would dictate that bass be at the upper end of this range by mid-December to increase their probability of overwinter survival. Young bass that are large enough to switch to a fish diet in early fall consume high-energy threadfin shad, a diet which accelerates their growth and enhances the probability that the bass achieve the critical size (about 25 ± 5 cm) for winter-spring survival. This critical survival size is determined primarily by the size of prey available in the spring; prey availability being dictated by winter water temperature. Thus, the size and abundance of available prey affects the amount of food consumed. Food consumption influences growth and growth in

turn dictates size of prey that can be consumed. Survival, therefore, depends on bass being able to grow large enough by the end of the growing season to store sufficient fat over the winter and to consume large shad in the spring.

IMPLICATIONS

Our studies indicate that in populations of predators where selective mortality of smaller fish is occurring, average growth of a year-class may be overestimated when back-calculated lengths are used. Using backcalculated lengths to estimate growth for preceding year-classes may also result in the erroneous conclusion that average growth has decreased from the oldest to the youngest year-class. This error may result in false inferences concerning the effects of various management practices or environmental alterations on growth and survival of a predator population. This size-selective mortality, which we see as a common but annually variable phenomenon in temperate reservoirs, would prevent development of generally applicable age-growth relationships as some authors have proposed. An understanding, however, of the effects of temperature on both predators and prey can aid in formulating management strategies for increasing predator production. Stroud (1967), in summarizing a symposium on reservoirs, emphasized the need for further knowledge about factors that control the availability and consumption of prey in order that dependable management techniques be developed.

Two management strategies for increasing the production of smaller bass would be possible: (1) increase the number of bass that reach the 25-cm critical size by the end of the growing season and (2) increase the availability of small prey in the spring. These strategies might be accomplished by managing the system to increase the vulnerability or standing stock of small prey which would be relatively resistant to cold mortality, such as small bluegill or a threadfin $\times$ gizzard shad hybrid (McLean et al. 1980). The carrying capacity of small centrarchids in a system may be increased by providing structural complexity such as brush piles (Cooper and Crowder 1979) that provide attachment space for fish food organisms, thus increasing bluegill production (Pardue 1973). Small centrarchids could be made more available to predators in the winter-early spring stress period by lowering water levels where brush piles and other structures would be exposed, thus increasing the vulnerability of these prey.

Reduced survival of smaller individuals in an age-class has implications for both the species and the ecosystem. A fewer number of individuals in an age-class could result in less intraspecific competition for food and space therefore resulting in higher growth rates of survivors. Also, interspecific competition may be reduced especially between young-of-the-year bass and other species. In an ecosystem such as Watts Reservoir where prey undergo large fluctuations in abundance, decreased survival of smaller predators may result in increased fitness to the population. With overwinter mortality of smaller fish, the ecosystem maintains only those larger fish thus maximizing the reproductive potential of the population.

This study has demonstrated that water temperature is a major factor controlling the structure of a predator population through its differential influence on abundance and size-

structure of prey. This seasonally fluctuating prey base decreases the survival and growth of age I⁺ and II⁺ largemouth bass smaller than 30 cm, regulating the structure of the population by controlling the number and sizes of individuals recruited into the adult segment of the population and influencing population reproductive potential.

Acknowledgments

We appreciate the assistance of J. A. Parrotta, J. Silver, P. A. Singley, and J. Hogue in collection and analysis of the data, and the helpful comments of C. C. Coutant, J. E. Breck, S. G. Hildebrand, W. Van Winkle, and S. J. Cavanaugh on the manuscript.

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